

Twitter Project API Query Options

This document outlines a few different options we have for running search queries. I ran a number of queries through the “recent search” API which pulls tweets from the past 7 days in reverse chronological order. For specific/narrow searches, I grabbed tweets from the entire week. For the broad ones, I restricted to a limited amount of time (usually a few hours). I then hand-coded all of the narrow search results, and took 300 tweet random samples of the broad searches. I then put together a table with a few metrics of how efficiently the queries pick up birth announcements and some very rough estimates of how many birth announcements we would get if we ran the code for the past year. Next, I discuss the syntax of queries briefly and provide a list of the queries I have put together. A few are missing from the table. I can always run those queries later if desired.

At this point, I think there are two major decisions to make:

1. Only use tweets with images, or use all original tweets¹
2. How broad we make our queries. This generally will involve a trade-off between the total volume of tweets we can pull in a given period of time, and the “hit-rate” (percent of pulled tweets that are birth announcements) So far, I’ve found this trade-off to be pretty steep.

For (1), I think we could get a sample of at least 3-4k image-only queries if we expanded the narrow queries some more and ran this over 5-6 years (say 2014-2019). However, if we’re interested in getting a larger sample, including tweets without pictures would be smart too. For (2), I think that the broad queries I have at this point are too broad. Maybe when we meet, we can brainstorm how to expand the narrow searches a little bit more. At the very least, they are a good starting point to build off of.

Performance

The table indicates query ID, hit rate, total number of tweets pulled from the API by that query, number of tweets I hand coded (sample size), number of tweets that are duplicated in another query, number of hours between the creation of the first tweet and the last tweet (not the run time for the query. Query runtime is very fast), estimated number of birth announcements for a year (yearly hits), estimated total tweets we would pull in a year, and what kind of restriction on media, images, links etc there are for the query². The table also includes totals of the yearly estimates of hits and total tweets for different categories of breadth and media requirements.

Based on my progress to this point, it is tricky to find a middle ground between very broad searches (thousands of tweets per *hour*, with $\approx 1\%$ hit rate) and fairly narrow searches (50-100 tweets per *week* with 10-25% hit rate). The higher come small samples so I wouldn’t necessarily bank on that high of a hit rate on average. I would imagine the narrow queries will at least be in the 10-15% range on average though, which is still pretty good.

The main limitation we will have is in the total number of tweets we can pull (it’ll probably be

¹For our purposes, “original tweet” excludes replies, retweets, and quote tweets.

²“No media” means we excluded tweets with images, videos, or links. “has images” means we excluded any tweet that did not have a picture.

Table 1: Query Performance Metrics

Query ID	Type	Total Tweets	Sample Size	Elapsed Hours	Duplicates	Hit Rate (%)	Yearly Hits	Yearly Tweets	Media Restriction
1	Narrow	43	43	158.62	5	13.95	312	2,236	Has images
2	Narrow	43	43	161.71	1	4.65	104	2,236	No media
3	Narrow	49	49	162.05	4	24.49	624	2,548	Has images
4	Narrow	121	121	162.68	2	6.61	416	6,292	No media
5	Narrow	26	26	159.88	4	26.92	364	1,352	Has images
6	Narrow	101	101	164.69	4	7.92	416	5,252	No media
7	Narrow	25	25	159.90	2	8.00	104	1,300	No links
8	Narrow	8	8	151.41	3	12.50	52	416	Has images
11	Broad	3,081	300	1.17	0	1.00	114,915	11,491,487	No restrictions
13	Narrow	28	28	1.95	10	0.00	0	62,809	No media
14	Broad	928	300	2.00	21	0.67	13,534	2,030,136	Has images
15	Broad	1,075	300	1.70	30	1.33	36,834	2,762,569	No restrictions
Total (Narrow)							2,340	21,216	
Total (Broad)							165,283	16,284,192	

Note: The total with images (Narrow) figure is the sum of queries 1,3,5,8. Grand total (narrow) is 1-8. Total (broad) is 11,14,15. The numbers that are missing did not end up getting run through the API. They really aren't that different from the rest, and would not change the general picture here.

about \$5,000 for every million tweets we pull). I don't think the number of queries we run or their level of sophistication will be an issue.

Syntax

Basically, each API query consists of a set of words and/or phrases. The API does exact matching on these strings. We can connect the strings with logic where "OR" does exactly what you would think while a space can be used for "AND" logic. However, we can nest quotation marks within the query to get an exact match for phrases with spaces in them. For example '("baby" "born") OR ("last night")' would pick up: "my baby was born last night", but would not pick up "last year, I had a baby born at night". All queries restrict to original tweets (no retweets, quote tweets, or replies). We can also toggle if there is media, images, videos, or links in the tweet. We can ignore tweets containing certain words by using "-" (e.g. -dog ignores any tweets containing "dog").

Query list

Note: I don't have data on 9,10, and 12. But I thought I would let you see them so that you can decide if you like any of the phrases/words included in them.

1. ("welcome to the world baby" OR "introducing baby" OR "our little one has arrived" OR "our new baby" OR "meet our baby" OR "our family just got bigger" OR "we had a baby" OR "our bundle of joy" OR "our baby is here" OR "we had our baby" OR ("born" "pounds" "ounces") OR ("born" "lbs" "oz")) has:images -is:retweet -is:quote -is:reply lang:en -puppy -cat -kitten -dog -"welcome to the world of" -"born again" -"chromosome" -"i was born"

2. ("welcome to the world baby" OR "introducing baby" OR "our little one has arrived" OR "our new baby" OR "meet our baby" OR "our family just got bigger" OR "we had a baby" OR "our bundle of joy" OR "our baby is here" OR "we had our baby" OR ("born" "pounds" "ounces") OR ("born" "lbs" "oz") OR "mom and baby are doing well") -has:media -is:retweet -is:quote -is:reply lang:en -puppy -cat -kitten -dog -"welcome to the world of" -"born again" -"chromosome" -"i was born"
3. ("just gave birth" OR "went into labor" OR "finally had my baby" OR "delivered naturally" OR "delivered vaginally" OR "gave birth naturally" OR "we had a baby girl" OR "we had a baby boy" OR "welcoming our newest blessing" OR "we became parents today" OR "our baby finally arrived" OR "we are overjoyed to announce the birth" OR "holding our baby for the first time" OR "our beautiful baby is finally here") has:images -is:retweet -is:quote -is:reply lang:en -puppy -cat -kitten -dog -is:retweet
4. ("just gave birth" OR "went into labor" OR "finally had my baby" OR "delivered naturally" OR "delivered vaginally" OR "gave birth naturally" OR "we had a baby girl" OR "we had a baby boy" OR "welcoming our newest blessing" OR "we became parents today" OR "our baby finally arrived" OR "we are overjoyed to announce the birth" OR "holding our baby for the first time" OR "our beautiful baby is finally here") -has:media -is:retweet -is:quote -is:reply lang:en -puppy -cat -kitten -dog -is:retweet
5. (("mom and baby are") OR ("i gave birth" "ago today") OR ("i had a baby") OR ("born" "spontaneous labor") OR ("born" "cesarean section") OR ("born" "c-section") OR ("had my baby today") OR ("i gave birth today") OR ("say hello to baby") OR ("say hi to baby")) has:images -is:retweet -is:quote -is:reply lang:en -puppy -cat -kitten -dog -dream -if
6. (("mom and baby are") OR ("i gave birth" "ago today") OR ("i had a baby") OR ("born" "spontaneous labor") OR ("born" "cesarean section") OR ("born" "c-section") OR ("had my baby today") OR ("i gave birth today") OR ("say hello to baby") OR ("say hi to baby")) -has:media -is:retweet -is:quote -is:reply lang:en -puppy -cat -kitten -dog -dream -if
7. (("born at" ":" "AM") OR ("born at" ":" "PM") OR "born at around" OR "born at exactly" OR ("our baby was born at" "this morning") OR ("our baby was born at" "last night")) -has:links -is:retweet -is:quote -is:reply lang:en -puppy -cat -kitten -dog -dream
8. (("born at" "AM") OR ("born at" "PM") OR ("born at around") OR ("born at exactly") OR ("our baby was born at" "this morning") OR ("our baby was born at" "last night")) has:images -is:retweet -is:quote -is:reply lang:en -puppy -cat -kitten -dog -dream
9. "was born early this morning" OR "meet your baby brother" OR "meet your baby sister" OR "welcome to the family" OR "new addition to the family" OR "was born late last night" OR "was born a few hours ago" OR "we are officially parents" has:images -is:retweet -is:quote -is:reply lang:en -puppy -cat -kitten -dog -dream

10. "was born early this morning" OR "meet your baby brother" OR "meet your baby sister"
OR "welcome to the family" OR "new addition to the family" OR "was born late last
night" OR "was born a few hours ago" OR "we are officially parents" -has:media
-is:retweet -is:quote -is:reply lang:en -puppy -cat -kitten -dog -dream
11. baby -is:retweet -is:reply -is:quote lang:en
12. ("gave birth" OR "had a baby" OR "had a child" OR "had a kid" OR "delivered our
baby" OR "welcomed our baby" OR "brought our baby into the world" OR "went into
labor" OR "born after a long labor") has:images -is:retweet -is:quote -is:reply
lang:en -puppy -cat -kitten -dog -dream
13. ("gave birth" OR "had a baby" OR "had a child" OR "had a kid" OR "delivered our
baby" OR "welcomed our baby" OR "brought our baby into the world" OR "went into
labor" OR "born after a long labor") -has:media -is:retweet -is:quote -is:reply
lang:en -puppy -cat -kitten -dog -dream
14. ("baby") has:images -is:reply -is:retweet -is:quote lang:en
15. (("baby" "my") OR ("child" "birth") OR ("gave birth") OR ("born" "today") OR ("born"
"last night") OR ("born" "yesterday") OR ("born" "ago today") OR ("delivered" "child")
OR ("delivered" "baby")) -is:retweet -is:quote -is:reply lang:en -puppy -cat -kitten
-dog -dream